SNOWPRO® SPECIALTY: GEN AI

EXAM STUDY GUIDE

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SNOWPRO® SPECIALTY: GEN AI STUDY GUIDE OVERVIEW

This is a self-paced guide that will outline the Snowflake domains, objectives, and topics that will be covered on the **SnowPro Specialty: Gen AI Certification Exam**. Use of this study guide does not guarantee certification success.

For an overview and more information on all of the available SnowPro Certifications, please navigate here.

RECOMMENDATIONS FOR USING THIS GUIDE

This guide outlines the Snowflake topics and subtopics covered on the exam. Following the topics will be additional resources consisting of links to documentation, blogs, and/or exercises that are designed to support candidates' understanding of Gen AI on Snowflake.

Estimated length of study time required to complete the guide: 10 – 13 hours

Some links may have more value for exam preparation than others, depending on the candidate's level of experience. The same amount of time should not be spent on each link. Some links may appear in more than one content domain.

SNOWPRO® SPECIALTY: GEN AI EXAM CONTENT AND FORMAT

CERTIFICATION OVERVIEW

The **SnowPro Specialty: Gen AI Certification Exam** tests specialized knowledge, skills, and best practices used to leverage Gen AI methodologies in Snowflake including key concepts, features, and programming constructs. The exam will assess skills through scenario-based questions, interactive questions, and real-world examples.

This certification will test the candidate's ability to:

- Define and implement Snowflake Gen AI principles, capabilities, and best practices related to infrastructure, data governance, and cost governance
- Leverage Snowflake Cortex AI features and functions, including Large Language Models (LLMs) to meet customer use cases (e.g., Cortex Analyst, Cortex Search, Cortex Code)
- Build and fine-tune open-source models with Snowpark Container Services and Snowflake Model Registry

The candidate is expected to have knowledge of:

- Snowflake Cortex suite of AI features and functions
- Snowpark Container Services and Snowflake Model Registry

Target Audience:

Candidates with one or more years of Gen AI experience with Snowflake, in an enterprise environment. In addition, successful candidates may have proficiency writing code in Python. Previous data engineering and SQL knowledge is assumed.

This exam is designed for:

- AI or ML Engineers
- Data Scientists
- Data Engineers
- Data Application Developers
- Data Analysts with programming experience

STEPS TO SUCCESS

1. Review this Gen AI Exam Study Guide
2. Attend Snowflake's Instructor-Led Snowflake Gen AI Training
3. Get hands-on experience with our Snowflake x Gen AI: A Cool Collab and Snowflake X Gen AI: LLM Functions
4. Review and study applicable white papers and documentation
5. Get hands-on practical experience with relevant business requirements using Snowflake
6. Attend free Snowflake Webinars
7. Attend Snowflake Virtual Hands-on Labs to gain practical experience
8. Practice with sample questions here
9. Schedule the exam
10. Take the exam!

EXAM DOMAINS AND WEIGHTINGS

The following table contains the domains and weightings covered on the exam. It is not a comprehensive listing of all the content that will be presented on the exam.

Domain	Domain Weightings
1.0 Snowflake for Gen AI Overview	18%
2.0 Snowflake Gen AI Functions	38%
3.0 Snowflake Gen AI Governance	29%
4.0 Snowflake Document Processing	15%

Domain 1.0: Snowflake for Gen AI Overview

1.1 Define Snowflake's Gen AI principles and features.

- Snowflake Cortex
    - Cortex Models and Functions
    - Cortex Fine-tuning (Public Preview)
    - Cortex Search
        - RAG use cases
        - Unstructured data use cases
    - Cortex Analyst
        - Text-to-SQL use cases
    - Cortex Agents
- Snowflake Cortex Code
- Cortex Code in Snowsight UI
    - Cortex Code Command Line (CLI)
- Snowflake Copilot Inline (Public Preview)
    - Cortex Models and Functions
    - Cortex Fine-tuning (Public Preview)
    - Cortex Search
        - RAG use cases
- Snowflake Intelligence
- Different interfaces
    - AI Studio
    - SQL
    - REST API
- Bringing your own models into Snowflake
    - Snowflake Model Registry (custom model)
    - Snowpark Container Services

1.2 Outline Gen AI capabilities in Snowflake.

- Prompting
- Cortex AI functions
    - Vector-embedding
    - Context Windows
- Cortex Search
    - Multi-index queries
    - Access control requirements
    - Different ways to use Cortex Search
- Cortex Analyst
    - Semantic Views
    - Semantic Views Autopilot
    - YAML Specification for Semantic Views
    - Verified Query
    - Custom Instructions
- Cortex Agents
- Snowflake Intelligence
- Cross-region inference
    - `CORTEX_ENABLED_CROSS_REGION` parameter
    - Considerations (e.g., latency, availability)
- REST APIs
- Model Context Protocol (MCP)
- Snowflake Cortex Code
    - Cortex Code CLI commands
- Cortex Knowledge Extensions (CKE)

Domain 1.0 Study Resources

Snowflake Documentation

Cortex Code in Snowsight
Cortex Code CLI Key Features
Cortex Fine-tuning
Snowflake Copilot Inline
Semantic Views Autopilot
YAML Specification for Semantic Views
Model Context Protocol (MCP) for Cortex Agents
Cortex Code CLI Session Management
Cortex Code CLI Commands
Cortex Knowledge Extensions Overview
Cross-Region Inference
Snowflake Cortex LLM Functions
Model Registry — Bring Your Own Model Types
Model Registry — Snowsight UI
Cortex Search Overview
Cortex Code CLI
Model Registry Overview
Cortex Analyst — Custom Instructions
Cortex Agents
Vector Embeddings
Snowflake ML Overview
Cortex AISQL — Model Access & Supported Features
Cortex Analyst REST API

Reading Assets

Cortex Search: AI Hybrid Search (Blog)
Agentic Semantic Model Text-to-SQL (Blog)

Domain 2.0: Snowflake Gen AI Functions

2.1 Apply AI functions in Snowflake.

- Snowflake Cortex AI functions
    - General
        - `AI_COMPLETE`
        - `COMPLETE Structured Outputs`
    - Task-specific functions
        - `AI_CLASSIFY`
        - `AI_EXTRACT`
        - `AI_PARSE_DOCUMENT`
        - `AI_SENTIMENT`
        - `SUMMARIZE`
        - `AI_SUMMARIZE_AGG`
        - `AI_TRANSLATE`
        - `AI_EMBED`
        - `AI_FILTER`
        - `AI_AGG`
        - `AI_SIMILARITY`
        - `AI_TRANSCRIBE`
        - `AI_REDACT`
    - Vector functions
        - `VECTOR_INNER_PRODUCT`
        - `VECTOR_L1_DISTANCE`
        - `VECTOR_L2_DISTANCE`
        - `VECTOR_COSINE_SIMILARITY`
        - `VECTOR_TRUNCATE`
        - `VECTOR_NORMALIZE`
        - `VECTOR_SUM`
        - `VECTOR_MIN`
        - `VECTOR_MAX`
        - `VECTOR_AVG`
    - Helper functions
        - `AI_COUNT_TOKENS`
        - `TRY_COMPLETE`
        - `SPLIT_TEXT_RECURSIVE_CHARACTER`
        - `SPLIT_TEXT_MARKDOWN_HEADER`
        - `TO_FILE`
        - `PROMPT`

2.2 Perform data analysis given a use case.

- Use fully-managed LLMs, RAG, and text-to-SQL services
    - Unstructured data
        - Functions
            - `AI_PARSE_DOCUMENT`
            - `AI_EXTRACT`
            - `AI_SIMILARITY`
            - `AI_COMPLETE`
        - Cortex Search
            - Recursive split text markdown
            - Chunk sizing
            - Embedding models
            - Semantic reranking
        - Multi-modal Analytics
            - Audio and Image Processing
    - Structured data
        - Functions
            - `AI_COMPLETE`
        - Cortex Analyst
            - Cortex Analyst Verified Query Repository (VQR)

- Integration with Cortex Search
- Suggested Questions
- `CUSTOM_INSTRUCTIONS`

- Performance considerations
  - Choosing a model
    - Latency (e.g., model size)
    - Accuracy (e.g., fine-tuning, reducing hallucinations)
    - Model capability
    - Provisioned Throughput

2.3 Build or interact with interfaces to chat with data in Snowflake.

- Set up the Snowflake environment
  - Required privileges
- Invoke Cortex functions within the application code (e.g., Streamlit in Snowflake)
  - Chat conversations
    - Multi-turn architecture
    - Update parameters (i.e., messages array for conversation history)
- Snowflake Intelligence

2.4 Apply Snowflake Cortex functions in data pipelines.

- Snowflake Cortex
  - SQL interface
  - Data extraction
  - Data enrichment
  - Data augmentation
  - Data transformations

2.5 Run third-party models in Snowflake.

- Using Snowpark Container Services
  - Environment setup
  - Docker images
  - Specification files
  - Create compute pool
  - Create image repository
- Using Snowflake Model Registry
  - Logging the model
  - Calling the model

Domain 2.0 Study Resources

Snowflake Documentation

Snowflake Cortex AI Functions (AISQL)
Provisioned Throughput
Choosing a Model
Getting Started with Generative AI in Snowflake
Snowflake Cortex LLM Functions
Cortex Guard
AI_CLASSIFY
Cortex Analyst Verified Query Repository
Vector Embeddings
VECTOR_INNER_PRODUCT
VECTOR_L1_DISTANCE
VECTOR_L2_DISTANCE
VECTOR_COSINE_SIMILARITY
Snowpark Container Services Overview
Model Registry Overview
SPCS Tutorial 1 — Create a Service
Cortex Analyst — Semantic Search Integration
EXTRACT_ANSWER (SNOWFLAKE.CORTEX)
COMPLETE (SNOWFLAKE.CORTEX)
Cortex Analyst
Cortex Analyst — Custom Instructions
Cortex Analyst — Custom Instructions (Advanced)
snowflake.cortex.complete (Python API)
snowflake.cortex.CompleteOptions (Python API)
AI_COMPLETE — Multimodal
Complete Structured Outputs
Cortex REST API — Embed
About Streamlit in Snowflake
AI_REDACT
Parsing Documents (AI_PARSE_DOCUMENT)
AI_EXTRACT
AI_SUMMARIZE_AGG
Native Apps with Containers (Snowflake CN)
AI_COUNT_TOKENS
AI_EMBED

Vector Data Types
Snowflake Intelligence
AI_PARSE_DOCUMENT
Document Extraction
BUILD_SCOPED_FILE_URL
AI_COMPLETE_SINGLE_FILE
SENTIMENT (SNOWFLAKE.CORTEX)
AI Documents
Cortex Agents — Manage

Quickstarts and Tutorials
Fine-Tuning an LLM in Snowpark Container Services with AutoTrain
Getting Started with Cortex Analyst (Quickstart)
Deploy & Fine-Tune LlamA 2 in SPCS (Quickstart)

Reading Assets
Snowpark Container Services — A Tech Primer (Blog)

Additional Resources
Northstar Developer Resources
Building Multimodal Data Pipelines Short Course

Domain 3.0: Snowflake Gen AI Governance

3.1 Set up model access controls.

- Limits on which models can be used
    - Restrict access to specific models
        - Application roles
    - Control model access
        - Role-Based Access Control (RBAC)
        - Account-level allowlist parameter
- Data safety and security considerations
    - Cross region inference
    - Guardrails
    - Sensitive data management (e.g., `AI_REDACT`)
    - Methods to reduce model hallucinations and bias
- REST API authentication methods

3.2 Grant and revoke Role-Based Access Control (RBAC) and privileges.

- Individual privileges
    - Specific requirements for Analyst, Search, Agents, and Snowflake Intelligence
- Roles
    - `CORTEX_USER`
    - `CORTEX_ANALYST_USER`
    - `CORTEX_AGENT_USER`
    - `CORTEX_EMBED_USER`

3.3 Manage, monitor, and optimize Snowflake Cortex costs.

- Cortex Agents
    - Limit token usage
- Cortex Search
    - Different types of costs (e.g., virtual warehouse, EMBED_TEXT, serving, indexing)
- Cortex Analyst
- Cortex AI functions
    - Minimize tokens
    - Token cost implications
- Tracking costs of Snowpark Container Services
    - Compute pools
- Tracking model usage and consumption
    - Usage quotas
    - `CORTEX_ANALYST_USAGE_HISTORY`
    - `CORTEX_AISQL_USAGE_HISTORY`
    - `CORTEX_SEARCH_DAILY_USAGE_HISTORY`
    - `CORTEX_REST_API_USAGE_HISTORY`
    - `CORTEX_PROVISIONED_THROUGHPUT_USAGE_HISTORY`
    - `METERING_DAILY_HISTORY`
    - `METERING_HISTORY`
- Object tagging to monitor AI services costs

3.4 Use Snowflake AI observability tools.

- Snowflake AI observability features
  - Evaluation metrics
  - Comparisons
  - Tracing
  - Logging
  - Event tables
- Implementation methods
  - Trulens SDK

Domain 3.0 Study Resources

Snowflake Documentation

Overview of Access Control
Governance Overview
Control Model Access
Cortex LLM Allowlist
Snowflake Intelligence
Cortex Analyst
CORTEX_FUNCTIONS_USAGE_HISTORY
COMPLETE (SNOWFLAKE.CORTEX)
COMPLETE (SNOWFLAKE.CORTEX) — Syntax
LLM Functions — Cortex Guard
snowflake.cortex.CompleteOptions (Python API)
Cortex Analyst — Admin Observability
AI Observability — Key Concepts
Snowflake Cortex LLM Functions
Cortex Search — Cost Categories
LLM Functions — Control Model Access
Cortex Search — Costs
LLM Functions — Required Privileges
AI Observability Reference — Data
AI Observability Reference
COMPLETE (SNOWFLAKE.CORTEX) — Arguments
CORTEX_SEARCH_DAILY_USAGE_HISTORY
Snowflake Cortex AISQL
Complete Structured Outputs — Cost Considerations
Model Registry — Cost Considerations
Opting Out — ACCOUNTADMIN and AI Features
Cortex Analyst — Cost Considerations
CORTEX_SEARCH_DAILY_USAGE_HISTORY — Columns
AI Observability Reference — Evaluation Metrics
ALTER CORTEX SEARCH
Cortex Agents
Arctic-Extract Fine-tuning
Snowflake Database Roles
AI Observability
Evaluate AI Applications
Redact PII
AI_REDACT
REVOKE PRIVILEGE on APPLICATION ROLE
Cortex Analyst — Required Privileges
LLM Functions — Limiting Access to Specific Roles
GRANT DATABASE ROLE
Cortex Agents — Cost Considerations

Reading Assets

Benchmarking LLM-as-a-Judge RAG Triad Metrics (Blog)
AI Observability: Evaluate Gen AI Apps with TruLens (Blog)
Cortex Search — Overview, Pricing & Cost Monitoring (Article)
Unlocking Privacy-First Intelligence with AI_REDACT (Blog)

Domain 4.0: Snowflake Document Processing

4.1 Use document parsing functions.

- `AI_PARSE_DOCUMENT`
  - `OCR mode`
  - `LAYOUT mode`
  - `page_split`
  - `page_limit`
- `AI_EXTRACT`
  - Response format
  - How to prompt/Prompt engineering

4.2 Prepare and manage documents and implement extracting workflows.

- Upload documents
- Requirements (e.g., formats, size limits)

4.3 Build automated document processing pipelines with Cortex AI integration.

- Orchestration of Snowflake tooling
  - Streams
  - Tasks

4.4 Troubleshoot and optimize document processing.

- Extracting query errors
  - `GET_PRESIGNED_URL` function
- Requirements and privileges
- Cost and best practice considerations
- Fine-tuning `arctic-extract` models

Domain 4.0 Study Resources

Snowflake Documentation

AI_EXTRACT
AI_PARSE_DOCUMENT
Working with Documents in Snowflake Cortex
Fine-tuning Arctic-Extract Models
AI_EMBED
Introduction to Streams
AI_PARSE_DOCUMENT — Arguments
Arctic-Extract Fine-tuning — Syntax
Cortex Fine-tuning
Introduction to Data Pipelines
Parsing Documents (AI_PARSE_DOCUMENT)
Document Extraction
Cortex AI Functions: Documents

Reading Assets

Ask Questions to Your Own Documents with Cortex Search (Developer Guide)

SNOWPRO® SPECIALTY: GEN AI SAMPLE QUESTIONS

1. A Gen AI Specialist needs to analyze the daily costs incurred for AI services in Snowflake.

   Which query will retrieve the credit consumption from Snowflake's metadata objects for data usage?

   A. `SELECT * FROM SNOWFLAKE.ACCOUNT_USAGE.QUERY_HISTORY WHERE SERVICE_TYPE='AI_SERVICES';`
   B. `SELECT * FROM SNOWFLAKE.INFORMATION_SCHEMA.METERING_HISTORY WHERE SERVICE_TYPE='AI_SERVICES';`
   C. `SELECT * FROM SNOWFLAKE.ACCOUNT_USAGE.METERING_HISTORY WHERE SERVICE_TYPE='AI_SERVICES';`
   D. `SELECT * FROM SNOWFLAKE.ACCOUNT_USAGE.METERING_DAILY_HISTORY WHERE SERVICE_TYPE='AI_SERVICES';`*

2. What is the primary role of memory in a multi-turn chat conversation using a Gen AI model in Snowflake Cortex Analyst?

   A. To securely store user credentials
   B. To increase the speed of response generation
   C. To maintain context throughout multiple requests*
   D. To limit the number of tokens processed for each request

3. A company processes multiple document types that require different extraction logic. Invoices need specific fields like vendor name, total amount, and line items extracted. Legal contracts need their full text converted to searchable Markdown with table structures preserved.

   How should the company structure their document processing workflow in Snowflake?

   A. Use `AI_PARSE_DOCUMENT` for all documents, then apply SQL filters to extract specific fields from invoices and preserve contract layouts
   B. Use `AI_EXTRACT` for all documents, defining separate response format schemas for invoices and contracts
   C. Use `AI_PARSE_DOCUMENT` in OCR mode for invoices to extract text, and `AI_EXTRACT` for contracts to pull the full document content
   D. Use `AI_EXTRACT` with a schema for invoices to pull structured fields, and `AI_PARSE_DOCUMENT` in `LAYOUT` mode for contracts to preserve document structure*

4. An organization needs to build a chatbot that answers customer queries by searching through unstructured documents stored as PDFs.

   Which Snowflake capability enables building this chatbot?

   A. Cortex Search*
   B. Cortex Analyst
   C. Cortex Agent
   D. Snowflake Intelligence

5. Which Snowflake Cortex LLM function should be used to generate an instructional lesson plan based on a prompt?

   A. `AI_COMPLETE`*
   B. `AI_EXTRACT`
   C. `SUMMARIZE`
   D. `AI_TRANSLATE`

MAINTAINING YOUR CERTIFICATION

All Snowflake Certifications expire two (2) years after your certification issue date.

SnowPro Certifications can now be recertified through the Snowflake Continuing Education (CE) program which includes these options -

- Completion of eligible Snowflake Instructor Led (ILT) Training Courses
- Earning of an equivalent or higher-level SnowPro Certification

Note: You must have a valid Certification to participate in the Continuing Education (CE) program.